Categorical predictors

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```r
library('tidyverse')
theme_set(theme_bw())
```

Categorical vs. numerical predictors

So far we have dealt with numerical predictors but often we are interested in differences between *groups* of some kind. How to deal with those? Fortunately, the linear modelling strategy we have learnt can easily cover these cases.

Two classes

Let's generate data for two groups (`A` and `B`), each with 30 data points, sampled from normal distributions with the means of 5 and 3 and the same standard deviation of 4:

```r
d <- tibble(score = c(rnorm(30, 5, 4), rnorm(30, 3, 4)),
            group = rep(c('A', 'B'), each = 30))
head(d)
```

```
## # A tibble: 6 x 2
##    score group
##    <dbl> <chr>
## 1  8.54  A
## 2  1.77  A
## 3  2.79  A
## 4  0.317 A
## 5  7.12  A
## 6 14.9   A
```

```r
tail(d)
```

```
## # A tibble: 6 x 2
##    score group
##    <dbl> <chr>
## 1  7.39  B
## 2  4.70  B
## 3  0.893 B
## 4 11.7   B
## 5 -1.35  B
## 6 -4.24  B
```

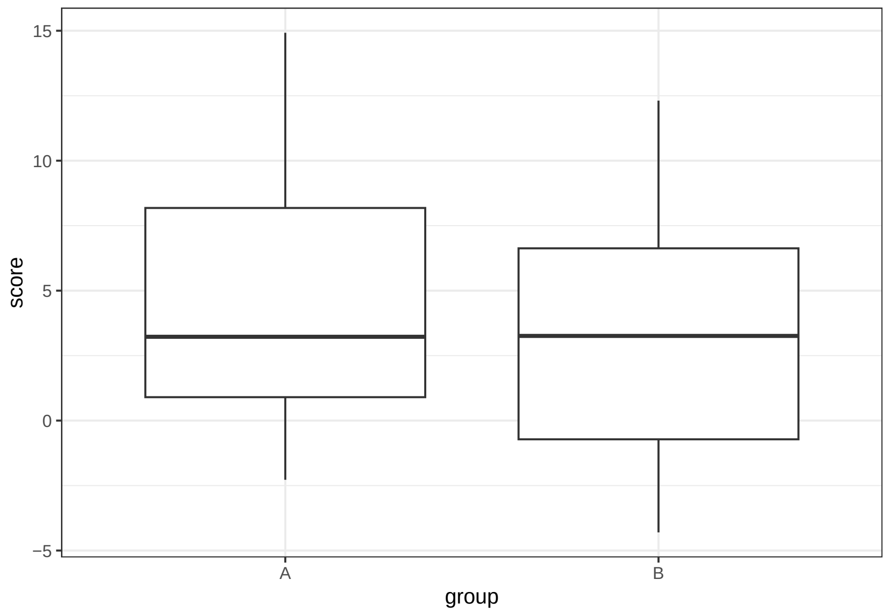
Since we `group` only takes one of two values (`A` or `B`), it makes sense to convert it to a *factor*:

```r
d <- d %>%
  mutate(group = as_factor(group))
d$group
```

```
##  [1] A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A B B B B B B B B
## [39] B B B B B B B B B B B B B B B B B B B B B B
## Levels: A B
```

We can plot this data as boxplots:

```r
ggplot(d, aes(x = group, y = score)) +
  geom_boxplot()
```



The question now is how we can *model* this data. As it happens, we can simply plug the categorical variable into the regression formula:

```r
m1 <- lm(score ~ group, d)
summary(m1)
```

```
##
## Call:
## lm(formula = score ~ group, data = d)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.2794 -3.6776 -0.5179  3.9103 10.4269
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.4943     0.8248   5.449 1.08e-06 ***
## groupB       -1.5158     1.1664  -1.299    0.199
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.518 on 58 degrees of freedom
## Multiple R-squared:  0.02829,    Adjusted R-squared:  0.01154
```

```
## F-statistic: 1.689 on 1 and 58 DF,  p-value: 0.1989
```

That's great but what do these coefficients mean? What R does under the hood is map one of the groups (by default the first one when group names are sorted alphabetically) to 0 (the intercept) and the other group to 1. Indeed we can see that:

- the intercept estimate (4.4943) is equal to the mean of group `A`
- the slope (-1.5158) corresponds to the difference between the second group mean and the intercept

```
d %>%
  group_by(group) %>%
  summarise(mean_score = mean(score))
```

```
## # A tibble: 2 x 2
##   group mean_score
##   <fct>      <dbl>
## 1 A           4.49
## 2 B           2.98
```

Note that we can *re-level* the group factor and make `B` the reference using `fct_relevel`:

```
d <- d %>%
  mutate(group = fct_relevel(group, 'B'))
levels(d$group)
```

```
## [1] "B" "A"
```

If we refit the model now, the intercept will be equal to the mean of group `B` and the slope will the same but with the opposite sign:

```
m2 <- lm(score ~ group, d)
coefficients(m2)
```

```
## (Intercept)      groupA
##    2.978518    1.515755
```

Exercise

The data set `ratings` from the `langaugeR` package includes, among other things, average ratings of referent size for a bunch of English words, marked for whether the referent is an animal or a plant:

```
library('languageR')
data('ratings')
```

Let's extract the `Word`, `Class` and `meanSizeRating` columns:

```
ratings <- ratings %>%
  select('Word', 'Class', 'meanSizeRating')
head(ratings)
```

```
##         Word  Class meanSizeRating
## 1     almond  plant         1.8912
## 2        ant animal         3.6275
## 3      apple  plant         2.4730
## 4    apricot  plant         1.7597
## 5 asparagus  plant         1.8660
## 6    avocado  plant         1.7737
```

1. Visualize the distribution of average size ratings for the animal and plant word categories and calculate their mean values.

```r
# Write your solution here.
```

2. Fit a regression model predicting mean size rating from the category of the referent and interpret the model coefficients.

```r
# Write your solution here.
```

3. Refit the model with "Plant" as the reference.

```r
# Write your solution here.
```

Three or more classes

The above can be easily generalized to situations when we have more than two classes:

```r
d3 <- tibble(score = c(rnorm(30, -5, 2), rnorm(30, 2, 2), rnorm(30, 8, 2)),
             group = rep(c('A', 'B', 'C'), each = 30))
```

Again, we will convert `group` to a factor:

```r
d3 <- d3 %>%
  mutate(group = as_factor(group))
```

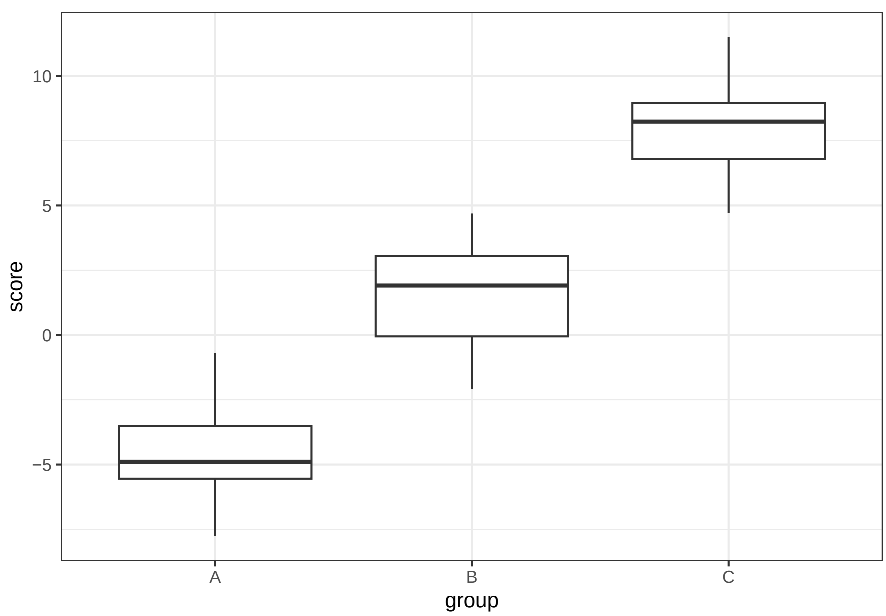
Note that since `A` comes first in the alphabet, it will be used as the reference:

```r
levels(d3$group)
```

```
## [1] "A" "B" "C"
```

Here is the distribution of `score` in each group:

```r
ggplot(d3, aes(group, score)) +
  geom_boxplot()
```



Let's fit the model:

```r
m3 <- lm(score ~ group, d3)
summary(m3)
```

```
##
## Call:
## lm(formula = score ~ group, data = d3)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -3.683 -1.316  0.150  1.332  3.959
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -4.6582     0.3426  -13.60   <2e-16 ***
## groupB        6.2468     0.4845   12.89   <2e-16 ***
## groupC       12.7246     0.4845   26.27   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.876 on 87 degrees of freedom
## Multiple R-squared:  0.888,  Adjusted R-squared:  0.8854
## F-statistic:   345 on 2 and 87 DF,  p-value: < 2.2e-16
```

As before, the intercept is equal to the mean of the reference group (in our case, `A`). The other two coefficients are equal to the difference between the respective group means and the intercept:

```
d3 %>%
  group_by(group) %>%
  summarise(mean_score = mean(score))
```

```
## # A tibble: 3 x 2
##   group mean_score
##   <fct>      <dbl>
## 1 A          -4.66
## 2 B           1.59
## 3 C           8.07
```

Note that we do not get the comparison between `B` and `C`. To do that we need to use one of them as a reference:

```
d3 <- d3 %>%
  mutate(group = fct_relevel(group, 'B'))
m4 <- lm(score ~ group, d3)
summary(m4)
```

```
##
## Call:
## lm(formula = score ~ group, data = d3)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -3.683 -1.316  0.150  1.332  3.959
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.5886     0.3426   4.637 1.24e-05 ***
## groupA       -6.2468     0.4845 -12.894  < 2e-16 ***
## groupC        6.4779     0.4845  13.371  < 2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.876 on 87 degrees of freedom
## Multiple R-squared:  0.888,  Adjusted R-squared:  0.8854
## F-statistic:   345 on 2 and 87 DF,  p-value: < 2.2e-16
```

Now we have the `B-C` and `B-A` comparisons but not the `A-C` comparison.

Exercise

Re-level the `group` factor to use `C` as a reference, refit the model and interpret the coefficients.

```
# Write your solution here.
```

Exercise

The file `lynott09.csv` contains a subset of data from Lynott, D. and Connell, L. (2009). *Behavior Research Methods* 41: 558. https://doi.org/10.3758/BRM.41.2.558, which measured the strength of association between adjectives and different modalities (auditory, haptic, olfactory etc.). **Remember to set the working directory to the location of the `Rmd` file before you attempt to read in the data.**

```
lynott09 <- read_csv('lynott09.csv')
```

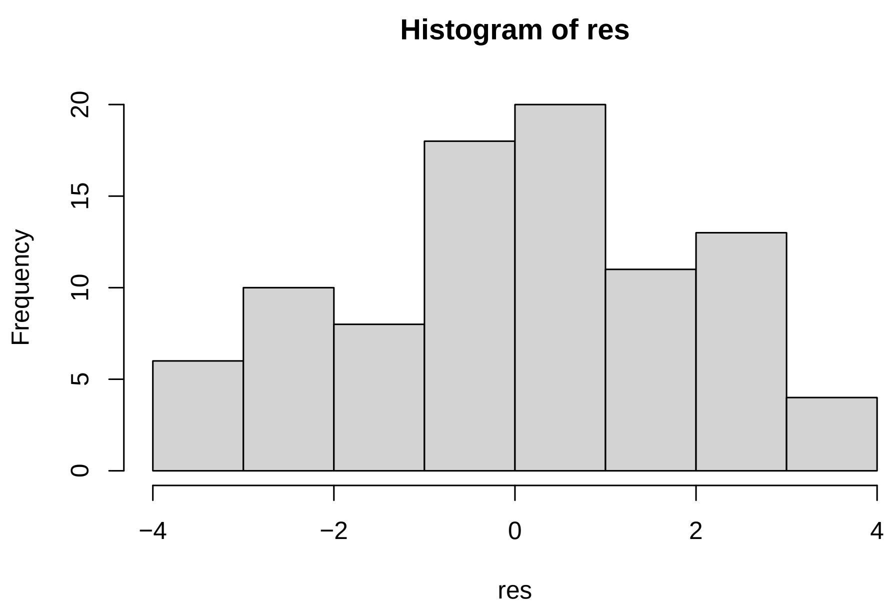
Create a model which predicts the degree to which adjectives with different dominant modalities (column `DominantModality`) are tied to a *single* perceptual modality (column `ModalityExclusivity`). Interpret the model coefficients.

```
# Write your solution here.
```

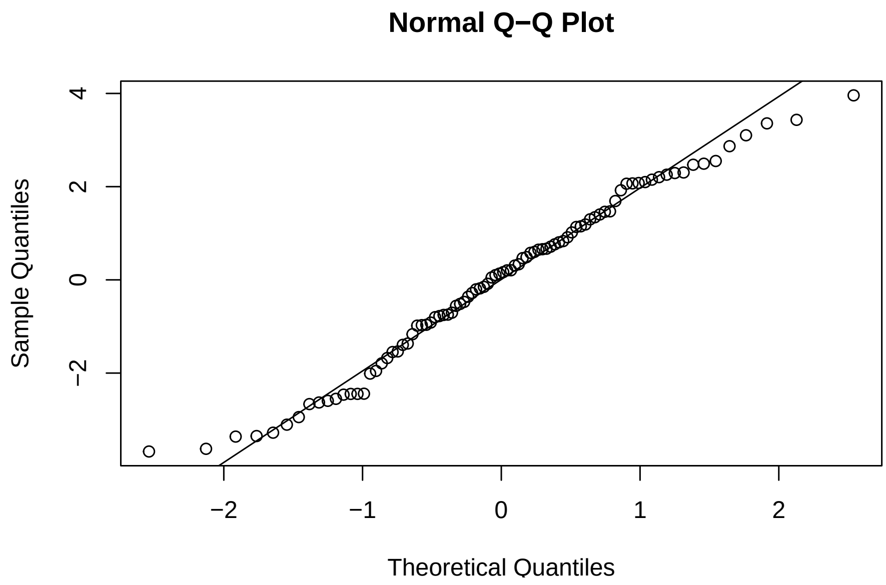
Revisiting the residuals

Remember analysis of residuals from last time. It is pretty much the same thing here. First of all, we would like residuals to be distributed normally.

```
res <- residuals(m4)
hist(res)
```

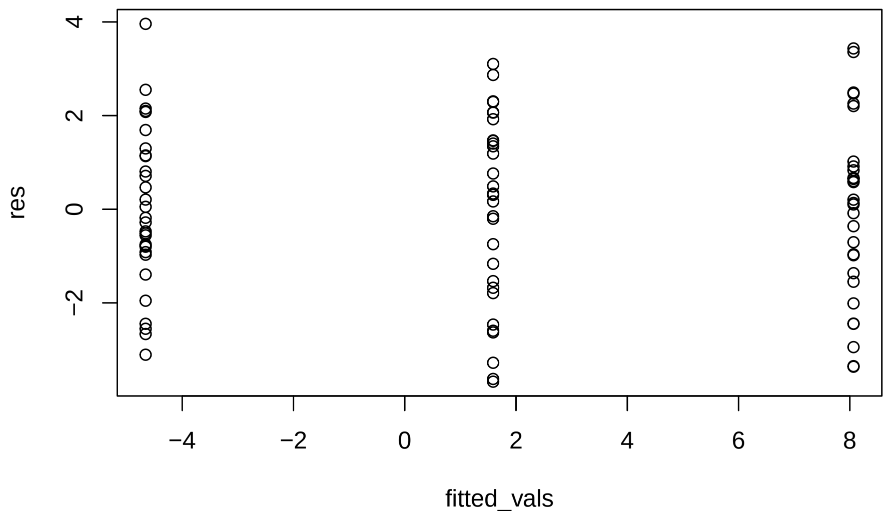


```r
qqnorm(res)
qqline(res)
```



Second, we want to see them have the same variance for each group:

```r
fitted_vals <- fitted(m4)
plot(fitted_vals, res)
```



That looks alright but are the fitted values? Remember that the expected value for each group is simply its mean:

```r
d3 %>%
  group_by(group) %>%
  summarise(mean_score = mean(score))
```

```
## # A tibble: 3 x 2
##   group mean_score
##   <fct>      <dbl>
## 1 B           1.59
## 2 A          -4.66
```

```
## 3 C           8.07
```

Remember how we said at the very beginning that you can think of the mean as a very simple model? Here you can see that this is really the case!

Exercise

Analyse the distribution of residuals in the `lynott09` model.

```
# Write your solution here.
```

Combining categorical and continuous predictors

Obviously, we can combine numerical and categorical predictors in a single model. Below we revisit the `lexdec` lexical decision data (from the `languageR` package, see `?lexdec` for more information) again:

```
data(lexdec)
```

Let's create a data summary with some statistics for each word:

- `Class`: category of the referent (`animal` or `plant`)
- `Frequency`: word frequency (log-transformed)
- `Length`: word length in characters
- `RT`: mean reaction time (in $\log(ms)$)

```
words <- lexdec %>%
  group_by(Word) %>%
  summarise(Class = first(Class),
            Frequency = first(Frequency),
            Length = first(Length),
            RT = mean(RT))
```

Let's fit a regression model which predicts reaction time (`RT`) from word frequency (`Freq`) and word class (`Class`)

```
lexdec_m <- lm(RT ~ Frequency + Class, words)
summary(lexdec_m)
```

```
##
## Call:
## lm(formula = RT ~ Frequency + Class, data = words)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.14003 -0.04067 -0.01793  0.04081  0.21881
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  6.640041   0.031403 211.445  < 2e-16 ***
## Frequency   -0.049164   0.005839  -8.420 1.73e-12 ***
## Classplant  -0.048229   0.014983  -3.219  0.00189 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06233 on 76 degrees of freedom
## Multiple R-squared:  0.4833, Adjusted R-squared:  0.4697
## F-statistic: 35.54 on 2 and 76 DF,  p-value: 1.272e-11
```

Here is how we would interpret this model (remember that reaction time is expressed on a logarithmic scale!):

- Intercept: the expected reaction time when both predictors are equal to 0, that is for a word denoting an animal with a frequency of 0. We can tell that the `animal` class was mapped on the intercept because `Classplant` is present in the model summary and therefore is not used as a reference. In addition, "animal" precedes "plant" when sorted in alphabetical order.
- Frequency: the change in reaction time corresponding to an increase of one in word frequency (note that frequency is also expressed on a logarithmic scale).
- Class: The difference in reaction time between the `plant` and `animal` categories.

Exercise

Using the data above, standardize `Frequency` and refit the model. How do you interpret the coefficients now?

```
# Write your solution here.
```

Exercise

Re-fit the model above with word length (`Length`) as an additional predictor. Interpret the coefficients.

```
# Write your solution here.
```